Experiment Set 3

Submission Due Date: 17/10/2026

Experiment 21

Using one 7-Segment LED Display, design a circuit using Arduino UNO and write one program to display numbers from 0 to 9. The display changes the number every two seconds continuously.

Answer Following Questions in your report..

Question 21.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 21.b: Explain how will you find out if the given 7-Segment LED is “Common Cathode” or “Common Anode” configuration.

Experiment 22

1. Use Program Titled “Experiment_22”

2. Connect the middle pin of potentiometer to A0 Analog Input of Arduino. Connect one pin of potentiometer to +5 V and other to Ground.

3. Open Serial Monitor from “Tools” Menu and see changes in values as you rotate the potentiometer switch.

4. Close “Serial Monitor” and open “Serial Plotter” and observe the change in resistance values being plotted.

```
/*
  AnalogReadSerial

  Reads an analog input on pin 0, prints the result to the Serial Monitor.
  Graphical representation is available using Serial Plotter (Tools > Serial Plotter menu).
  Attach the center pin of a potentiometer to pin A0, and the outside pins to +5V and ground.

  This example code is in the public domain.

  http://www.arduino.cc/en/Tutorial/AnalogReadSerial
*/

// the setup routine runs once when you press reset:
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
}

// the loop routine runs over and over again forever:
void loop() {
  // read the input on analog pin 0:
  int sensorValue = analogRead(A0);
  // print out the value you read:
  Serial.println(sensorValue);
  delay(1);        // delay in between reads for stability
}
```

Answer Following Questions in your report..

Question 22.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 22.b: Explain following functions.

(22.b.1) Serial.begin(9600);

(22.b.2) Serial.println(..)

Experiment 23

1. Use Program Titled “Experiment_23”

2. Connect a push-button switch in series with resistor. Connect +5 V to other end of resistor and connect the other terminal of switch to Ground. Connect the junction of Switch and Resistor to Pin 2 of Arduino.

3. Observe the serial monitor. Observe what happens when you press switch. Does value in Serial monitor change?

4. Observe similar change in serial plotter.

```
/*
  DigitalReadSerial

  Reads a digital input on pin 2, prints the result to the Serial Monitor

  This example code is in the public domain.

  http://www.arduino.cc/en/Tutorial/DigitalReadSerial
*/

// digital pin 2 has a pushbutton attached to it. Give it a name:
int pushButton = 2;

// the setup routine runs once when you press reset:
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
  // make the pushbutton's pin an input:
  pinMode(pushButton, INPUT);
}

// the loop routine runs over and over again forever:
void loop() {
  // read the input pin:
  int buttonState = digitalRead(pushButton);
  // print out the state of the button:
  Serial.println(buttonState);
  delay(1);        // delay in between reads for stability
}
```

Answer Following Questions in your report..

Question 23.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Experiment 24

1. Use Program Titled “Experiment_24”
2. Connect LED to Pin 9 of Arduino, in series with a resistor.
3. Connect a potentiometer, middle pin to A0 pin, side pins to +5 V and Ground.
4. Observe the output on Serial Monitor and Plotter. Observe the change in intensity of LED.

```
/*
  Analog input, analog output, serial output

  Reads an analog input pin, maps the result to a range from 0 to 255 and uses
  the result to set the pulse width modulation (PWM) of an output pin.
  Also prints the results to the Serial Monitor.

  The circuit:
  - potentiometer connected to analog pin 0.
    Center pin of the potentiometer goes to the analog pin.
    side pins of the potentiometer go to +5V and ground
  - LED connected from digital pin 9 to ground

  created 29 Dec. 2008
  modified 9 Apr 2012
  by Tom Igoe

  This example code is in the public domain.

  http://www.arduino.cc/en/Tutorial/AnalogInOutSerial
*/

// These constants won't change. They're used to give names to the pins used:
const int analogInPin = A0;  // Analog input pin that the potentiometer is attached to
const int analogOutPin = 9; // Analog output pin that the LED is attached to

int sensorValue = 0;        // value read from the pot
int outputValue = 0;        // value output to the PWM (analog out)

void setup() {
  // initialize serial communications at 9600 bps:
  Serial.begin(9600);
}

void loop() {
  // read the analog in value:
  sensorValue = analogRead(analogInPin);
  // map it to the range of the analog out:
  outputValue = map(sensorValue, 0, 1023, 0, 255);
  // change the analog out value:
  analogWrite(analogOutPin, outputValue);

  // print the results to the Serial Monitor:
  Serial.print("sensor = ");
  Serial.print(sensorValue);
  Serial.print("\t output = ");
  Serial.println(outputValue);

  // wait 2 milliseconds before the next loop for the analog-to-digital
  // converter to settle after the last reading:
  delay(2);
}
```

Answer Following Questions in your report..

Question 24.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 24.b: Explain following functions.

(24.b.1) map(sensorValue, 0, 1023, 0, 255)

(24.b.2) Serial.print("\t output = ");

(24.b.3) Serial.println(outputValue);

Experiment 25

1. Use Program Titled “Experiment_24”
2. Connect LED to Pin 9 of Arduino, in series with a resistor.
3. Connect a potentiometer, middle pin to A0 pin, side pins to +5 V and Ground.
4. Change the Experiment 24 by sending output without using “map” function.

Answer Following Questions in your report..

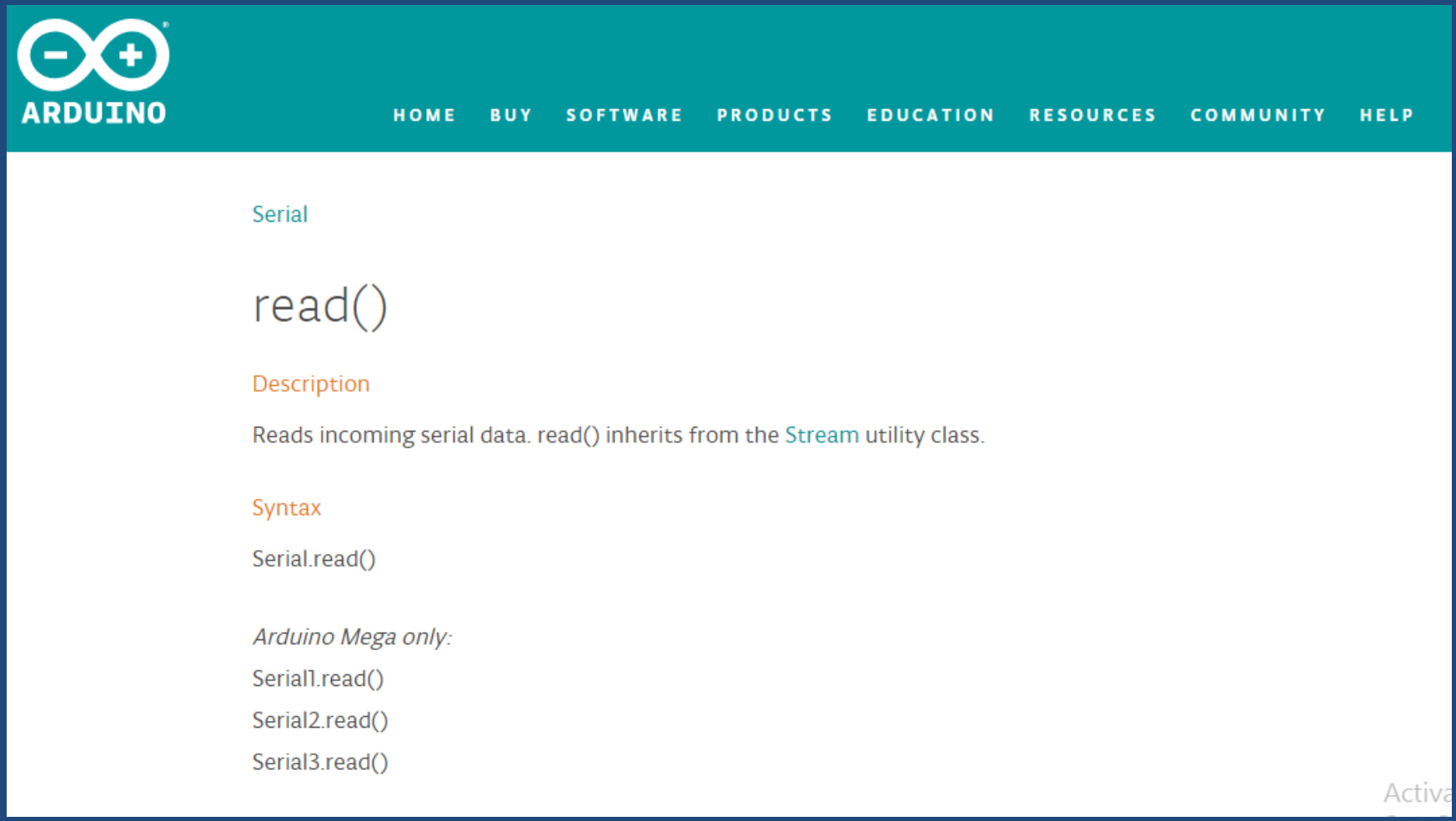
Question 25.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Experiment 26

1. Use Program Titled “Experiment_26”. Update and modify the program if needed.

2. Understand and execute the program.

Serial Read


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Serial
read()
Description
Reads incoming serial data. read() inherits from the Stream utility class.
Syntax
Serial.read()
Arduino Mega only:
Serial1.read()
Serial2.read()
Serial3.read()

Parameters

None

Returns

the first byte of incoming serial data available (or -1 if no data is available) - *int*

Example

```
int incomingByte = 0;   // for incoming serial data

void setup() {
        Serial.begin(9600);     // opens serial port, sets data rate to 9600 bps
}

void loop() {

        // send data only when you receive data:
        if (Serial.available() > 0) {
                // read the incoming byte:
                incomingByte = Serial.read();

                // say what you got:
                Serial.print("I received: ");
                Serial.println(incomingByte, DEC);
        }
}
```

Answer Following Questions in your report..

Question 26.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 26.b: Explain following functions.

(26.b.1) Serial.available( )

(26.b.2) Serial.read( )

Experiment 27

1. Modify the program and hardware of Experiment 26 in the following way.

2. Read an integer number between 3 and 5 from serial monitor input.

3. If the number is out of range, then print on serial monitor "Number is out of Range."

4. If number is "3", turn on LED on Pin 3 for 2 seconds.
   If number is "4", turn on LED on Pin 4 for 2 seconds.
   If number is "5", turn on LED on Pin 5 for 2 seconds.

Answer Following Questions in your report..

Question 27.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 27.b: Include your new C Program in your report.

Question 27.c: Describe the changes you have made in the program.

Distance Measurement using Ultrasonic Sensor

Experiment 28

1. Connect the Ultrasonic Sensor to Arduino.

2. Use Program Titled “Experiment_28”. Update and modify the program if needed.

3. Understand the program and perform the experiment.

HC-SR04 Ultrasonic Sensor



Ultrasonic Sensor Pin Configuration

Pin Number	Pin Name	Description
1	Vcc	The Vcc pin powers the sensor, typically with +5V
2	Trigger	Trigger pin is an Input pin. This pin has to be kept high for 10us to initialize measurement by sending US wave.
3	Echo	Echo pin is an Output pin. This pin goes high for a period of time which will be equal to the time taken for the US wave to return back to the sensor.
4	Ground	This pin is connected to the Ground of the system.

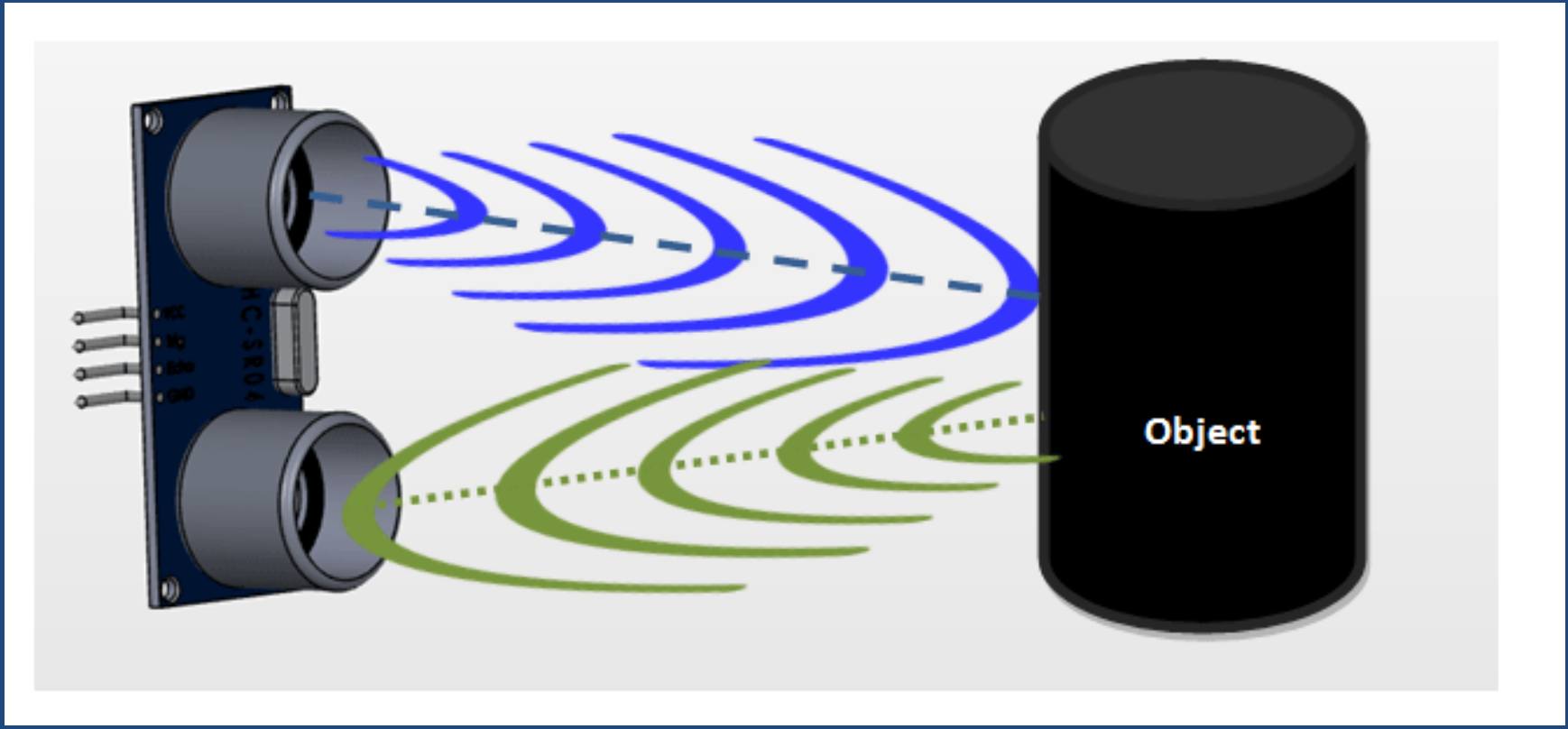
- Source: https://components101.com/ultrasonic-sensor-working-pinout-datasheet

HC-SR04 Sensor Features

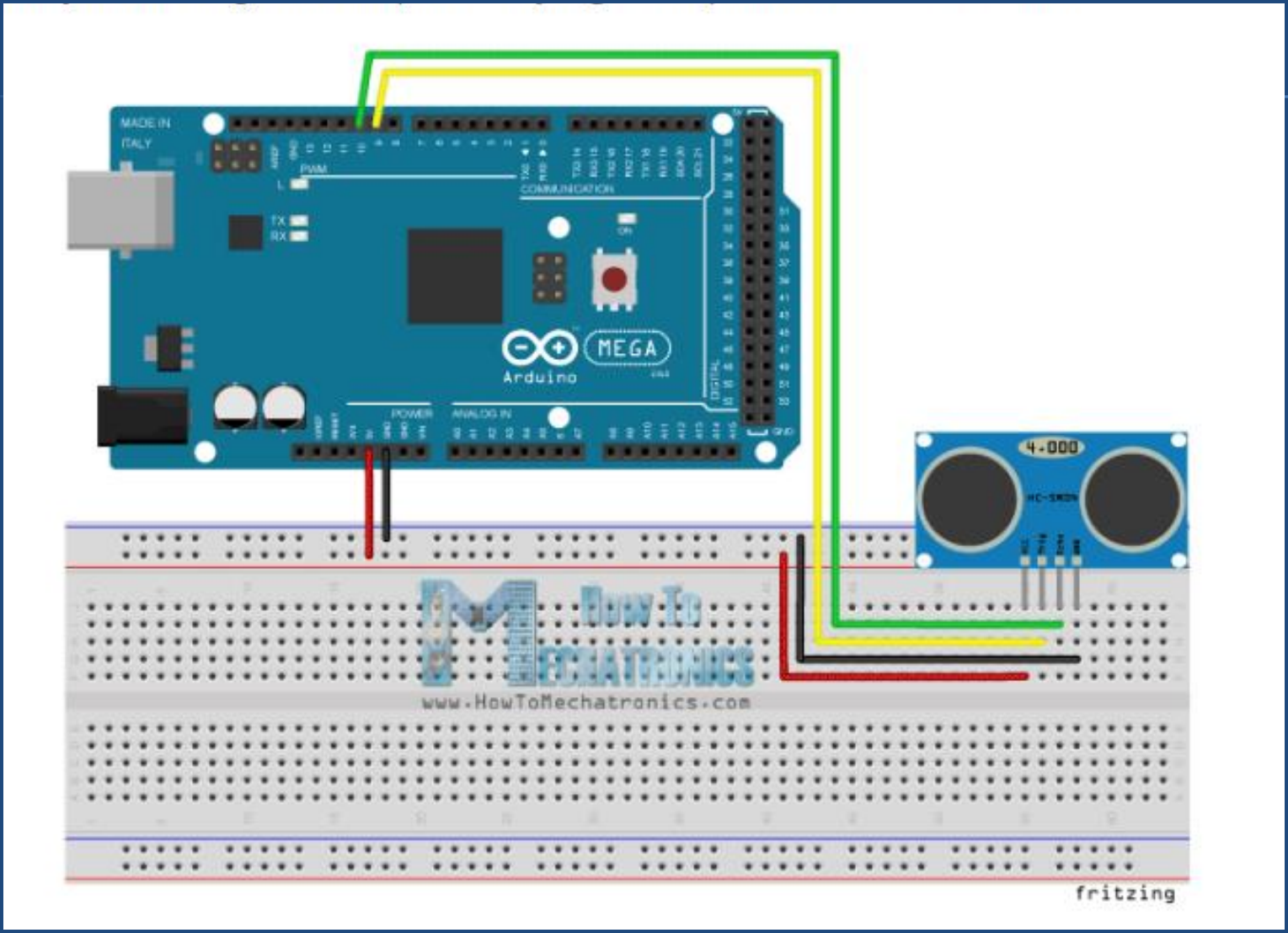
- Operating voltage: +5V
- Theoretical Measuring Distance: 2cm to 450cm
- Practical Measuring Distance: 2cm to 80cm
- Accuracy: 3mm
- Measuring angle covered: <15°
- Operating Current: <15mA
- Operating Frequency: 40Hz

- Frequency : 40 KHz (40,000 Hz)

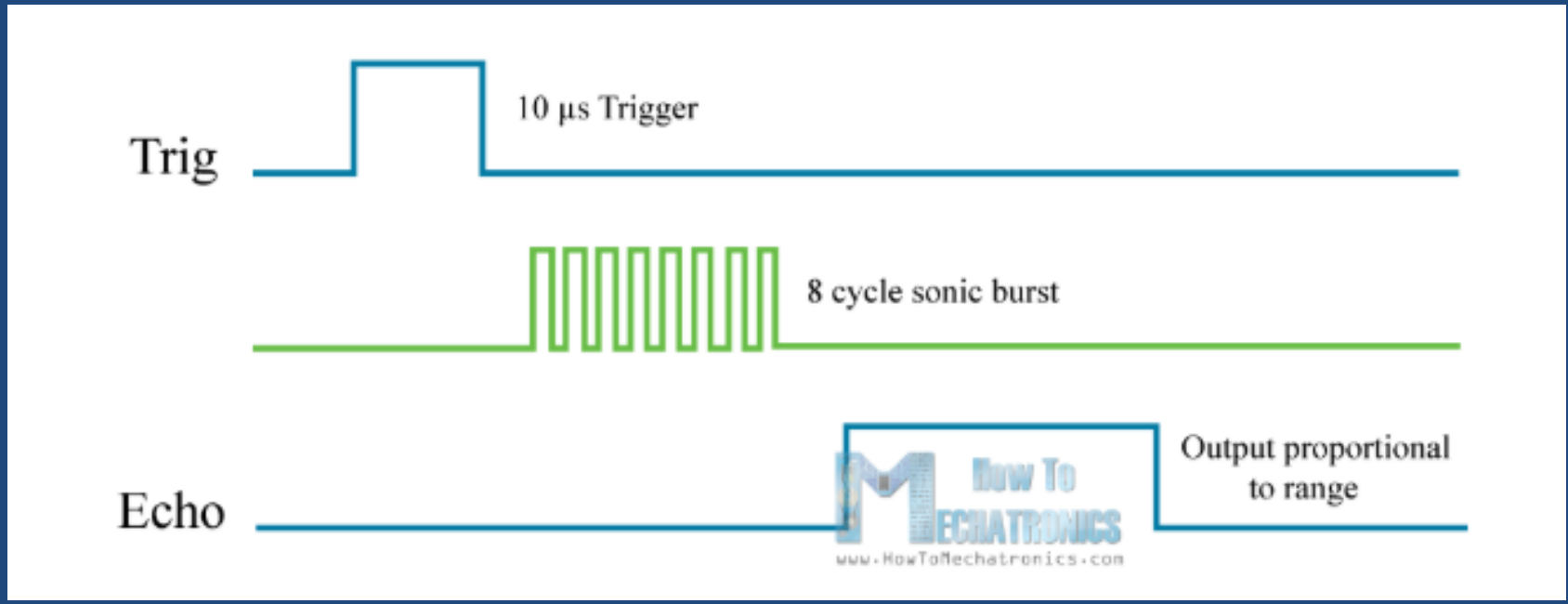
HOW SENSOR WORKS ?



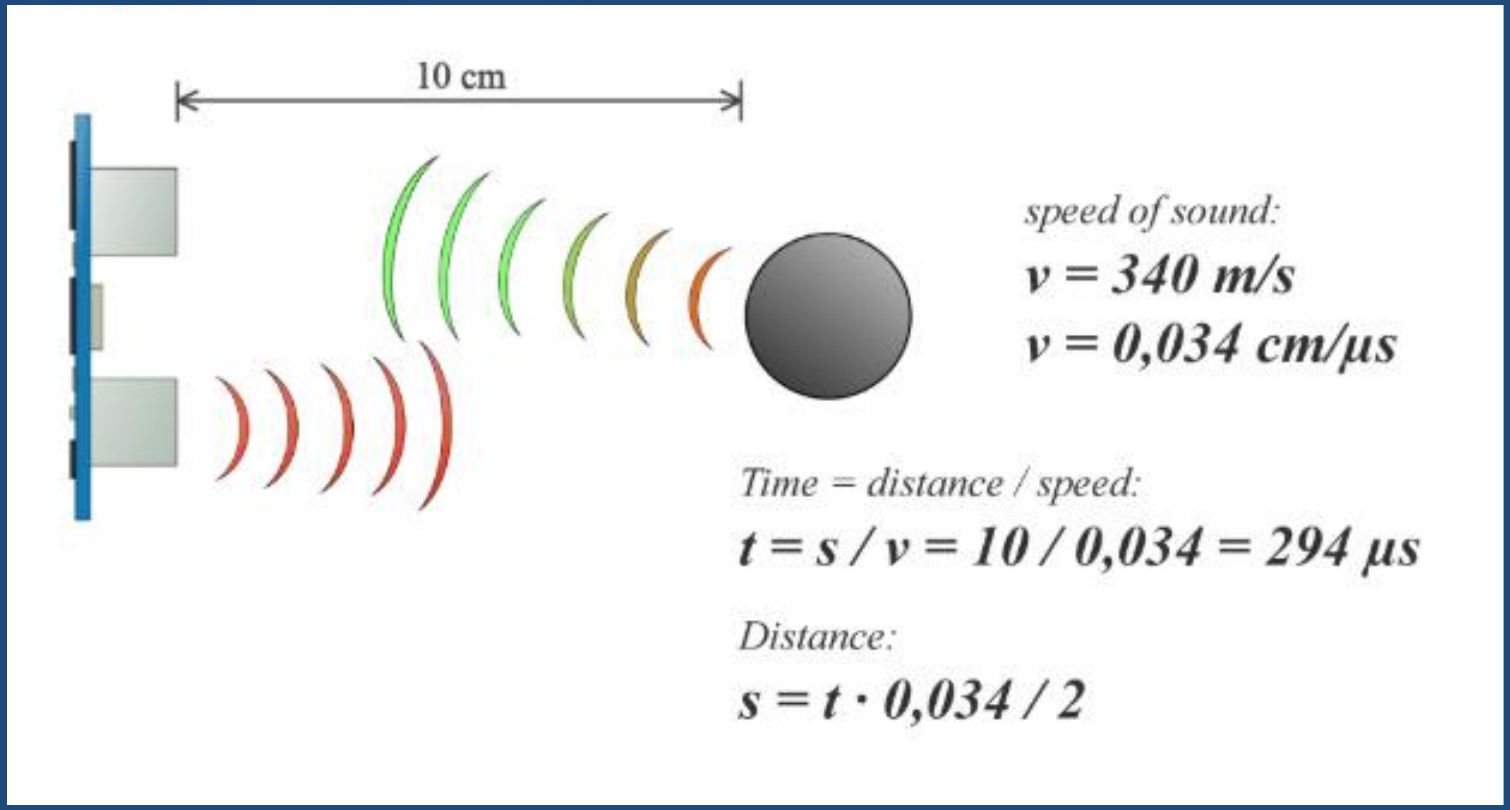
Source: https://components101.com/ultrasonic-sensor-working-pinout-datasheet



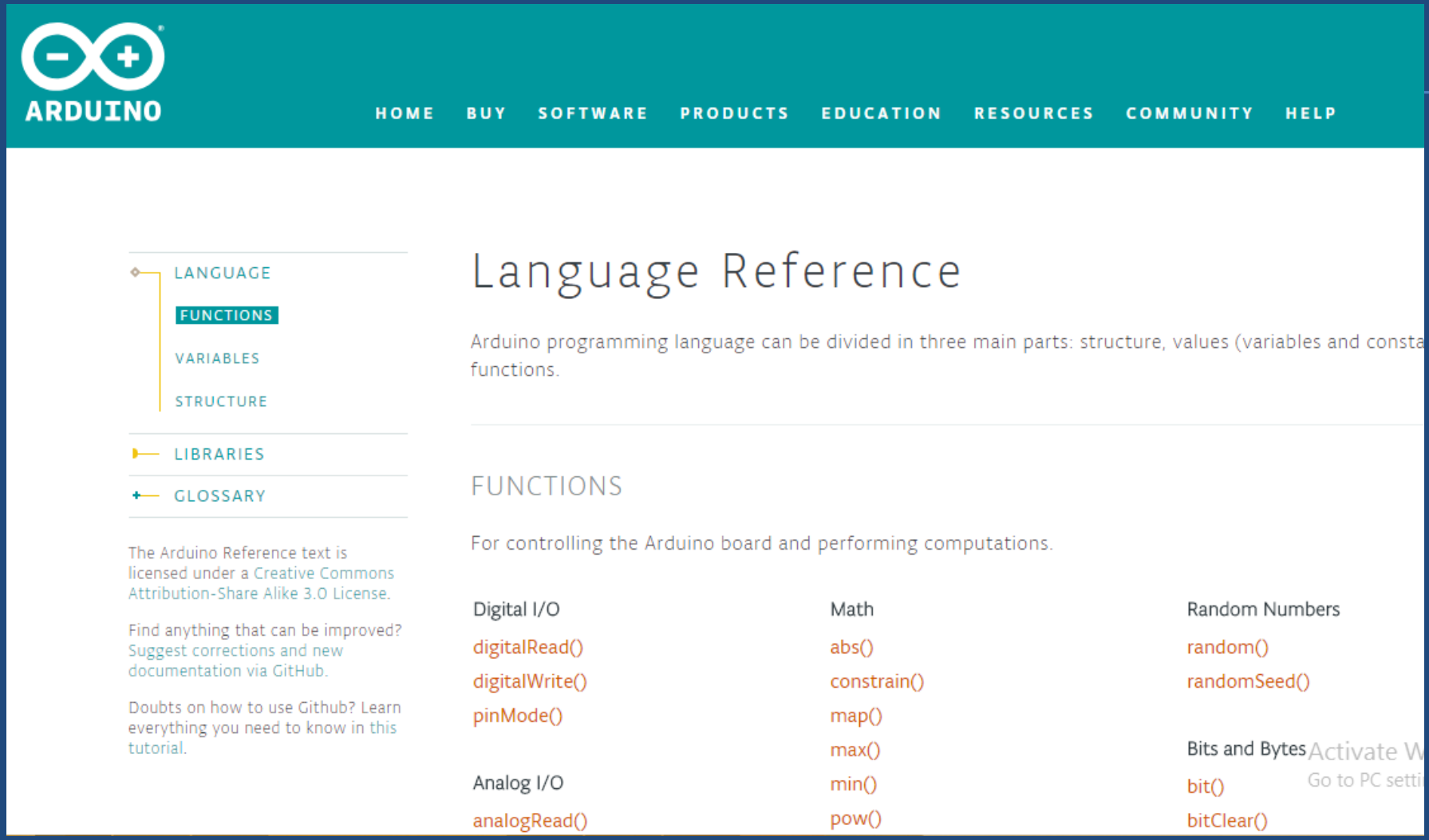
Source: https://components101.com/ultrasonic-sensor-working-pinout-datasheet



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Find anything that can be improved? Suggest corrections and new documentation via GitHub.
Doubts on how to use Github? Learn everything you need to know in this tutorial.
Language Reference
Arduino programming language can be divided in three main parts: structure, values (variables and consta
functions.
FUNCTIONS
For controlling the Arduino board and performing computations.
Digital I/O
digitalRead()
digitalWrite()
pinMode()
Analog I/O
analogRead()
Math
abs()
constrain()
map()
max()
min()
pow()
Random Numbers
random()
randomSeed()
Bits and Bytes
bit()
bitClear()
Activate W
Go to PC setti

Reference > Language > Functions > Advanced io > Pulsein

pulseIn()

[Advanced I/O]

Description

Reads a pulse (either `HIGH` or `LOW`) on a pin. For example, if `value` is `HIGH`, `pulseIn()` waits for the pin to go from `LOW` to `HIGH`, starts timing, then waits for the pin to go `LOW` and stops timing. Returns the length of the pulse in microseconds or gives up and returns 0 if no complete pulse was received within the timeout.

The timing of this function has been determined empirically and will probably show errors in longer pulses. Works on pulses from 10 microseconds to 3 minutes in length.

Syntax

```
pulseIn(pin, value)

pulseIn(pin, value, timeout)
```

Parameters

`pin`: the number of the pin on which you want to read the pulse. (int)

`value`: type of pulse to read: either HIGH or LOW. (int)

`timeout` (optional): the number of microseconds to wait for the pulse to start; default is one second (unsigned long)

Returns

the length of the pulse (in microseconds) or 0 if no pulse started before the timeout (unsigned long)

Example Code

The example calculated the time duration of a pulse on pin 7.

```
int pin = 7;
unsigned long duration;

void setup()
{
  pinMode(pin, INPUT);
}

void loop()
{
  duration = pulseIn(pin, HIGH);
}
```

Activate Windows
Go to PC settings to activate

Answer Following Questions in your report..

Question 28.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 28.b: Explain following function.

- duration = pulseIn(echoPin, HIGH)

Question 28.c: Explain the equation used to calculate distance using the ultrasonic sensor. How will you change this formula if you do the measurement under the water?

Experiment 29

1. Modify the program of experiment 28 in the following way.

2. Continuously send the data to the website “ThingSpeak.com”. Observe the data and Graph.

Answer Following Questions in your report..

Question 29.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 29.b: Include your new c program in your report.

Question 29.c: Describe the changes you have made in the program.

Experiment 30

1. Use Program Titled “Experiment_30”. Update and modify the program if needed.

2. Perform above experiment and observe the result on LCD.

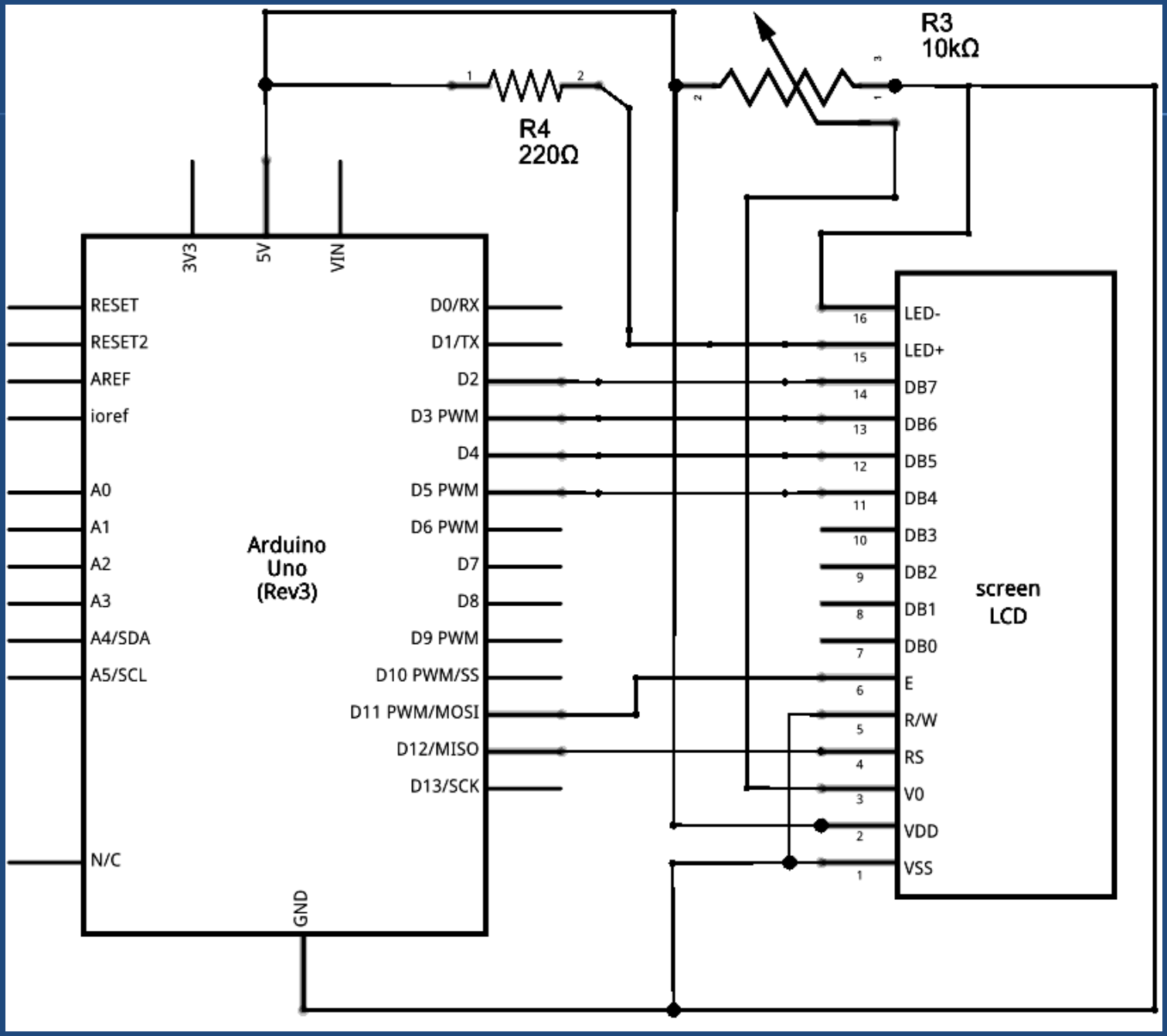
3. Modify the Program to change the Text you are sending to display with today’s date and observe the output.

Answer Following Questions in your report..

Question 30.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report. Prepare a combined video for Point 1 and 3.

Question 30.b: Include your new c program in your report that you have updated for Point 3.

Question 30.c: Describe the changes you have made in the program.


R3
10kΩ
R4
220Ω
3V3
5V
VIN
RESET
RESET2
AREF
ioref
A0
A1
A2
A3
A4/SDA
A5/SCL
N/C
GND
D0/RX
D1/TX
D2
D3 PWM
D4
D5 PWM
D6 PWM
D7
D8
D9 PWM
D10 PWM/SS
D11 PWM/MOSI
D12/MISO
D13/SCK
Arduino
Uno
(Rev3)
LED-
LED+
DB7
DB6
DB5
DB4
DB3
DB2
DB1
DB0
E
R/W
RS
V0
VDD
VSS
screen
LCD

Experiment 31

1. Use Program Titled “Experiment_31”. Update and modify the program if needed.

2. Get the data from the server built through another Arduino (Like Experiment 18 and 19).

Answer Following Questions in your report..

Question 31.a: Prepare a video recording with appropriate voice over explaining the execution of this program. Save the video in any folder/drive and include the link to that video in this report.

Question 31.b: Study the program and answer following questions.

(31.b.1) Which library files are included ?

(31.b.2) How “mac address” is specified ?

(31.b.3) How “IP address” is specified ?

(31.b.4) How data are transferred on internet ?